

BANARAS HINDU UNIVERSITY

B.Com. (Hons.) — Semester IV (Special) Examination, 2014-15

Paper No. BCH-226: Business Statistics

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer all questions. All questions carry equal marks (14 marks each).

Question 1. Explain the procedure of constructing a cost of living index number. What are the difficulties associated with it? [14]

OR

Define Index Number and mention its uses and limitations. [14]

Question 2. Differentiate between the Fixed Base and Chain Base methods of constructing index numbers and discuss their relative merits. [14]

OR

Calculate the real wage index numbers from the data given below. How much increase in money wages is required to maintain the real wages intact at the 2000 level?

Year	2000	2001	2002	2003	2004	2005	2006
Wages (in Rs. 00)	20	24	35	36	36	38	40
Price Index	100	150	200	220	230	250	250

[14]

Question 3. What are the components of a time series? How would you find out the trend values in a time series by the method of least squares? [14]

OR

Find out the value of the trend by three-yearly moving averages and plot them on a graph paper:

Year	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004
Value	10	15	12	18	15	22	19	24	20	26

[14]

Question 4. What is seasonal variation in a time series? Describe the different methods you know to evaluate it and examine their relative merits. [14]

OR

Fit a parabolic curve of second degree ($y = a + bx + cx^2$) to the production data given below:

Year	2000	2001	2002	2003	2004
Production (000 tonnes)	8	9	10	7	5

[14]

Question 5. What do you understand by Statistical Quality Control? What are its advantages and disadvantages? [14]

OR

Write short notes on the following:

- (a) Control Charts
- (b) Process Control and Product Control

[14]